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Valve: Research

Transcatheter vs Surgical Aortic Valve Replacement in Bicuspid Aortic Valves

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ABSTRACT

BACKGROUND Recent approval of transcatheter aortic valve replacement (TAVR) in patients at lower risk profiles has resulted in a real-world expansion in patients with bicuspid aortic valves (BAV), otherwise excluded from trials comparing TAVR with surgical aortic valve replacement (SAVR). This study compared perioperative and longitudinal outcomes between BAV patients undergoing TAVR vs SAVR.

METHODS Using the United States Centers for Medicare and Medicaid Services inpatient claims database, we evaluated all beneficiaries with BAV undergoing isolated SAVR or TAVR (2018-2022). Comorbidities and frailty were accounted for using validated metrics with doubly robust risk adjustment using inverse probability weighting, multilevel regression models, and competing-risk time to event analysis. Subgroup analysis evaluated patients <75 years with low surgical risk (<4%).

RESULTS The study included 11,289 BAV patients (8123 SAVR and 3166 TAVR). Accounting for age, comorbidities, and frailty, TAVR was associated with lower procedural mortality (odds ratio, 0.40; $P < .001$) but higher pacemaker (12.4% vs 2.3%; odds ratio, 5.4; $P < .001$), longitudinal stroke (2.4% vs 1.5%; hazard ratio [HR], 1.35; $P < .001$), and all-cause mortality (8.8% vs 5.7%; HR, 1.49; $P < .001$) compared with SAVR. The young low-risk subgroup (5393 SAVR and 1731 TAVR) highlighted similar findings, with TAVR associated with higher longitudinal stroke (2.1% vs 1.7%; HR, 1.22; $P = .017$) and composite stroke, valve reintervention, or death (8.1% vs 5.9%; HR, 1.37; $P < .001$) compared with SAVR.

CONCLUSIONS Among Medicare beneficiaries with BAV, TAVR was associated with lower index in-hospital mortality but also lower 5-year risk-adjusted freedom from longitudinal stroke compared with SAVR, even in the youngest low-risk patients.

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Recent approval of transcatheter aortic valve replacement (TAVR) in patients at low surgical risk has resulted in a rapid expansion of TAVR in all cohorts.[1-4] As a result, the Society of Thoracic Surgeons (STS) predicted risk of mortality may no longer be as highly considered by the heart team when deciding treatment allocation.[5-8] This recent change to clinical practice has resulted in TAVR being used in populations not examined by low-risk trials comparing TAVR against surgical aortic valve replacement (SAVR). These include younger patients, patients with aortic

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Abbreviations and Acronyms

AKI = acute kidney injury
BAV = bicuspid aortic valve
CMS = Centers for Medicare and Medicaid Services
HR = hazard ratio
ICD-10 = International Classification of Diseases, 10th Revision
NOTION-2 = Comparison of TAVR With SAVR in Younger Low Surgical Risk Patients With Severe Aortic Stenosis
OR = odds ratio
SAVR = surgical aortic valve replacement
STS = The Society of Thoracic Surgeons
TAVR = transcatheter aortic valve replacement
PVL = perivalvular Leak

insufficiency, and those with bicuspid aortic valves (BAV).[9-11] Importantly, BAVs were specifically excluded from all prior TAVR trials because generally younger patients, oval geometry, and higher calcium scores in these valves were thought to be less favorable for TAVR.[12,13]

Recent data from the Comparison of TAVR With SAVR in Younger Low Surgical Risk Patients With Severe Aortic Stenosis (NOTION-2) trial highlight a higher composite end point of stroke or death in BAV patients treated with TAVR compared with SAVR, but portions of this trial were underpowered.[14] These data may be furthered by an examination of a larger real-world population. Additionally, recent longitudinal evidence from the STS Adult Cardiac Surgery Database has highlighted excellent long-term outcomes after SAVR with 95% survival at 8 years in low-risk patients including those with BAV.[9,15] In combination with the highly selected and relatively small cohorts evaluated in prospective trials, these data highlight the potential unique anatomic and clinical outcome differences in this population as well as the need for larger comparisons of TAVR vs SAVR in the BAV population to inform current clinical decision-making and to further inform the design of future trials in this population.

The purpose of the current analysis was to evaluate real-world contemporary longitudinal outcomes of TAVR vs SAVR in Medicare beneficiaries with BAV stenosis. We hypothesized that SAVR would be associated with higher early morbidity and mortality but superior longitudinal freedom from death, stroke, or valve reintervention over the study period.

MATERIAL AND METHODS

PATIENT DATA. Using the United States Centers for Medicare and Medicaid Services (CMS) inpatient claims database, we evaluated all beneficiaries aged 65 to 85 years with documented BAV undergoing isolated SAVR (n = 8123) or TAVR (n = 3166) between January 1, 2018, and December 31, 2022. We excluded patients with concomitant valvular procedures, coronary procedures, aortic procedures, prior heart surgery or valve procedure, endocarditis, pure aortic insufficiency, mechanical support, cardiogenic shock, or emergency procedures (Supplemental Table 1). To specifically address potential clinical decision bias within the national claims' data set, patients aged >85 years were excluded a priori based on lack of clinical relevance of SAVR vs TAVR and valve durability, and a subgroup analysis further evaluated low-risk patients aged <75 years. The West Virginia University Institutional Review Board approved this study with waiver of consent (Protocol #2210660362, 3/23/23).

VARIABLES AND DEFINITIONS. Procedures were identified through diagnosis-related group and International Classification of Diseases, 10th Revision (ICD-10) Procedure Coding System codes. Variable adjudication followed standard methodology of ICD-10 coding (Supplemental Table 2).[16] The Elixhauser comorbidity index, including 30 comorbidity variables, was used as a composite measure of baseline health using ICD-10 Clinical Modification codes.[17] The Kim index, a validated claims-based measurement of frailty, was used for risk adjustment using 52 ICD-10 diagnosis code-based variables, 25 Current Procedural Terminology (American Medical Association) code-based variables, and 16 Healthcare Common Procedure Coding System code-based variables in the past 12 months to predict the value of a deficit-accumulation frailty index.[18]

A previously validated predicted risk score for isolated SAVR using CMS data similar to the STS predicted risk of mortality was used to identify low-risk patients (Supplemental Methods, Supplemental Table 3).[19] Present on admission modifiers were used to delineate preoperative risk factors vs postoperative complications. Longitudinal data were obtained from CMS claims data for the primary indication for subsequent hospital admissions. Heart failure readmission was assessed with a 30-day blanking period to account for the differences in perioperative course. There were no missing data.

STATISTICAL ANALYSIS. Outcomes included index hospitalization mortality or complications, including acute kidney injury (AKI), new stroke, major bleeding, surgical repair of the femoral

TABLE 1 Baseline Demographics, Comorbidities, and Outcomes

Variable	Total (N = 11,289)	TAVR (n = 3166)	SAVR (n = 8123)	*P*
Age, y	70 (66–75)	71 (67–75)	69 (66–72)	<.001
Male sex	7284 (64.5)	1829 (57.8)	5455 (67.2)	<.001
White race	10,200 (90.4)	2855 (90.2)	7345 (90.4)	.135
Frailty index	0.155 ± 0.04	0.164 ± 0.044	0.152 ± 0.037	<.001
Elixhauser score	5.38 ± 1.96	5.04 ± 1.98	5.51 ± 1.94	<.001
Diabetes mellitus	2457 (21.8)	851 (26.9)	1606 (19.8)	<.001
Peripheral artery disease	1007 (8.92)	402 (12.7)	605 (7.45)	<.001
Chronic lung disease	2191 (19.4)	750 (23.7)	1441 (17.7)	<.001
Gastrointestinal bleeding, ulcer, reflux	2870 (25.4)	789 (24.9)	2081 (25.6)	.459
Liver disease	452 (4.0)	173 (5.46)	279 (3.43)	<.001
Atrial fibrillation	2458 (21.8)	712 (22.5)	1746 (21.5)	.261
Coronary artery disease	4750 (42.1)	1646 (52)	3104 (38.2)	<.001
Anemia	1054 (9.34)	412 (13)	642 (7.9)	<.001
Cancer	403 (3.57)	181 (5.72)	222 (2.73)	<.001
Cerebrovascular accident	234 (2.0)	63 (2.0)	171 (2.1)	.237
Permanent pacemaker	724 (6.4)	237 (7.5)	487 (6.0)	<.001
Heart failure	4215 (37.3)	1960 (61.9)	2255 (27.8)	<.001
Obesity	2734 (24.2)	730 (23.1)	2004 (24.7)	.071
Malnutrition	332 (2.94)	96 (3.03)	236 (2.91)	.767
Chronic renal disease	1859 (16.5)	683 (21.6)	1176 (14.5)	<.001
Surgical predicted risk of mortality, %	1.8 (0.9–3.2)	1.9 (0.9–3.1)	1.7 (0.8–3.3)	.038
Hospital				
Length of stay, d	6 (2–8)	2 (1–5)	8 (4–10)	<.001
Total cost, $	35,755 ± 24,911	39,168 ± 26,156	34,425 ± 24,281	<.001
Acute kidney injury	1413 (12.5)	197 (6.22)	1216 (15.0)	<.001
Stroke	216 (1.88)	62 (1.96)	146 (1.80)	.086
Bleeding	1546 (13.7)	419 (13.2)	1127 (13.9)	.456
New pacemaker	483 (4.3)	293 (12.4)	190 (2.34)	<.001
Vascular complication	21 (0.2)	13 (0.4)	8 (0.1)	<.001
Death	120 (1.06)	17 (0.537)	103 (1.27)	<.001
30 days				
Readmit	1093 (9.68)	239 (7.55)	854 (10.5)	<.001
Readmit bleeding	23 (0.204)	1 (0.032)	22 (0.271)	.020
Readmit stroke	43 (0.381)	17 (0.537)	16 (0.19)	.056
Readmit heart failure	487 (4.31)	117 (3.7)	470 (5.55)	.049
Any death	213 (1.89)	47 (1.48)	166 (2.04)	.067
5 years				
Any death	697 (6.17)	260 (8.21)	437 (5.38)	<.001
Readmit	3574 (31.7)	1050 (33.2)	2524 (31.1)	.034
Readmit bleeding	75 (0.664)	6 (0.19)	69 (0.849)	.001
Readmit stroke	226 (2.0)	88 (2.8)	138 (1.7)	<.001
Readmit heart failure	1530 (13.6)	557 (17.6)	973 (12)	<.001
Valve reintervention	54 (0.478)	12 (0.379)	42 (0.517)	.422
Composite[a]	800 (7.09)	325 (10.3)	475 (5.85)	<.001

[a]Composite: death, valve reintervention, or stroke. Data are presented as n (%), mean ± SD, or median (quartile 1–quartile 3). SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement.

artery, and new pacemaker placement. Longitudinal outcomes included all-cause mortality, readmission for stroke, readmission for valve reintervention (rereplacement or balloon valvuloplasty), readmission for bleeding related to anticoagulation, and readmission for heart failure. The primary outcome was the composite longitudinal death, stroke, or valve reintervention.

Categorical variables are presented as n (%), and continuous variables were assessed for normality and are presented as mean (SD) or median (quartile 1-quartile 3), as appropriate. Univariable comparisons were performed using χ^2

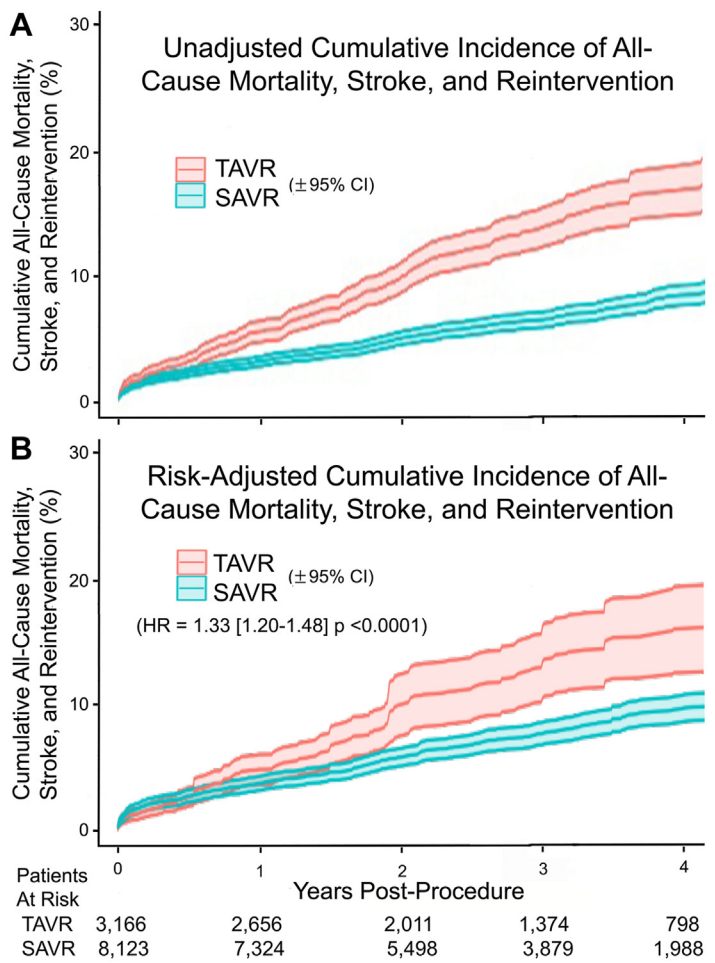

FIGURE 1 Cumulative incidence of all-cause mortality, stroke, and valve reintervention. (A) Unadjusted cumulative incidence and (B) risk-adjusted cumulative incidence for composite primary outcome, surgical aortic valve replacement (SAVR) vs transcatheter aortic valve replacement (TAVR) with 95% CI (shaded area) (hazard ratio [HR], 1.33; $P < .001$).

for categorical comparisons, and nonparametric tests assessed continuous variables given non-normal distribution.

To account for confounding in this retrospective analysis, we used inverse probability weighting of propensity scores. The inverse probability of treatment weighting models included all preoperative demographic, comorbidity, and frailty variables (Table 1). An acceptable balance threshold was set a priori at a standard mean difference of <0.10, with good balance between groups (Supplemental Figure).

Doubly robust risk adjustment was performed using multilevel regression analysis, Cox proportional hazards survival modeling, and Fine-Gray time-to-event analysis accounting for competing risk of death. All baseline covariates were included, and models were assessed for collinearity using the variance inflation factor. Longitudinal outcomes were evaluated to 4 years with censoring. Subgroup analysis was performed in similar fashion for low-risk patients, aged <75 years, excluding patients with surgical predicated risk of mortality of >4%.

RESULTS

PATIENT CHARACTERISTICS AND UNADJUSTED OUTCOMES. A total of 11,289 BAV patients (8123 SAVR and 3166 TAVR), most of whom were low surgical risk (predicted risk <4%), were analyzed with a median follow-up time of 2.6 and 2.4 years, respectively. Before risk adjustment, the median age was 70 years with 64.5% men. Patients undergoing TAVR had a higher frailty index but lower Elixhauser score, with no difference in race or prior stroke (Table 1). Isolated SAVR patients had lower rates of concomitant coronary artery disease (38% vs 52%, $P < .0001$). The predicated risk of index hospitalization mortality for isolated surgical aortic valve replacement was higher in the TAVR patients compared with SAVR (1.7% vs 1.9%, $P = .038$).

Evaluating postoperative outcomes, patients receiving TAVR had lower rates of AKI (6.2% vs 15.0%, $P < .0001$) but higher rates of new pacemaker insertion (12.4% vs 2.3%, $P < .0001$) (Table 1). Although there was no difference in index stroke, when considering index and readmission for stroke within 30 days of discharge, TAVR patients had significantly more new strokes (2.0% vs 2.5%, $P = .021$). Patients undergoing SAVR had a longer median hospital length of stay (8 days vs 2 days, $P < .0001$) but lower total cost ($34,425 vs $39,168, $P < .0001$) (Table 1). Index mortality was higher in SAVR (1.3% vs 0.5%, $P < .0001$), but no difference was observed between SAVR vs TAVR when index and 30 days after discharge were combined (2.0% vs 1.5%, $P = .067$).

TAVR patients had higher all cause readmission over the 5-year study period (31.1% vs 33.2%, $P = .034$), driven primarily by higher readmission for stroke (1.7% vs 2.8%, $P < .0001$) and heart failure (12.0% vs 17.6%, $P < .0001$) (Table 1). Unadjusted time-to-event analysis demonstrated higher overall survival ($P < .0001$) and event-free survival ($P < .0001$) (Figure 1A) for SAVR compared with TAVR.

RISK-ADJUSTED OUTCOMES. After comprehensive risk adjustment accounting for age, comorbidities,

TABLE 2 Risk-Adjusted Outcomes

Variable	Hazards Ratio for TAVR (95% CI)	*P* Value
Hospital		
Acute kidney injury[a]	0.385 (0.349–0.423)	.001
Stroke[a]	1.78 (1.49–2.14)	.001
Bleeding[a]	0.98 (0.85–1.12)	.783
Vascular complication[a]	3.6 (1.15–14.6)	.041
New permanent pacemaker[a]	1.6 (1.37–1.88)	.001
Death[a]	0.75 (0.579–0.98)	.038
30 days		
Readmit[a]	0.53 (0.48–0.59)	.001
Readmit bleeding[a]	0.13 (0.06–0.24)	.001
Readmit stroke[a]	1.29 (0.84–2.01)	.246
Readmit heart failure[a]	0.47 (0.41–0.55)	.001
5 years		
Death[b]	1.49 (1.16–1.92)	.002
Readmit[c]	0.99 (0.88–1.12)	.913
Readmit bleeding[c]	0.178 (0.10–0.29)	.001
Readmit stroke[c]	1.31 (1.07–1.59)	.008
Readmit heart failure[c]	1.23 (1.01–1.51)	.045
Valve reintervention[c]	0.64 (0.396–1.03)	.072
Composite[c,d]	1.33 (1.2–1.48)	.001

[a]Multivariable logistic regression; [b]Cox proportional hazards model; [c]Fine-Gray multivariable time-to-event model accounting for competing risk of death; [d]Composite: death, valve reintervention, or stroke. TAVR, transcatheter aortic valve replacement.

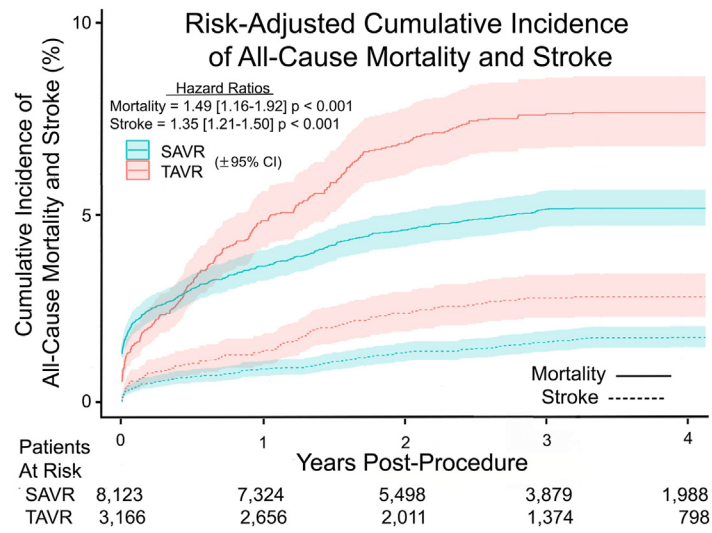

FIGURE 2 Risk-adjusted cumulative incidence of mortality and stroke for surgical aortic valve replacement (SAVR) vs transcatheter aortic valve replacement (TAVR) with 95% CI (shaded areas) ($P < .001$).

including concomitant coronary disease, and frailty, TAVR was associated with lower procedural mortality (odds ratio [OR], 0.40; $P < .001$) but higher stroke (OR, 1.78; $P < .0001$), new pacemaker (OR, 1.60; $P < .001$), and access complications (OR, 3.6; $P = .041$) compared with SAVR (Table 2). For the primary outcome, risk-adjusted longitudinal analysis demonstrated that TAVR was associated with a higher composite of death, stroke, or valve reintervention compared with SAVR (14.3% vs 7.9%; hazard ratio [HR], 1.33; $P < .001$) (Figure 1B). For the secondary outcomes, TAVR was associated with higher risk-adjusted longitudinal all-cause mortality (8.8% vs 5.7%; HR, 1.49; $P < 0.001$) and stroke (2.8% vs 1.7%; HR, 1.35; $P < 0.001$) (Figure 2), compared with SAVR. Accounting for the 30-day blanking period, SAVR was associated with greater freedom from longitudinal readmission for heart failure (HR, 1.23; $P = .045$)

LOW-RISK SUBANALYSIS, AGE <75 YEARS. A total of 7124 patients with BAV aged <75 years were identified as low risk (5393 SAVR vs 1731 TAVR). The median predicated risk score was <1% and similar between SAVR and TAVR groups (0.66% vs 0.68%, $P = .614$) (Table 3). Before risk adjustment, TAVR patients had higher rates of new pacemaker (1.5% vs 6.6%, $P <$ .001) but lower risk of major bleeding (3.8% vs 1.9%, $P < .001$) and AKI (12.1% vs 4.4%, $P < .001$), with no difference in perioperative stroke (1.5% vs 1.6%, $P = .821$) or hospital death (0.5% vs 0.2%, $P = .077$). Unadjusted longitudinal analysis highlighted that SAVR patients had lower rates of all-cause mortality (3.2% vs 4.8%, $P = .002$), readmission for heart failure (9.7% vs 12.7%, $P <$ .001), and readmission for stroke (1.6% vs 2.3%, $P = .045$). The unadjusted composite end point of stroke, death, or valve reintervention occurred more commonly in the TAVR patients (3.5% vs 6.6%, $P < .001$) (Figure 3A).

After doubly robust risk adjustment, TAVR compared with SAVR in low-risk patients with BAV aged <75 years was associated with lower AKI (OR, 0.41; $P < .001$) but higher rates of new pacemaker (OR, 1.62; $P < .001$) (Table 4). Longitudinal risk-adjusted analysis in the younger low-risk BAV subgroup demonstrated TAVR was associated with higher longitudinal stroke (1.7% vs 2.1%; HR, 1.22; $P = .017$) compared with SAVR, as well as the composite primary end point of stroke, death, or valve reintervention (4.7% vs 6.4%; HR, 1.37; $P < .001$) (Figure 3B).

COMMENT

This contemporary real-world examination of aortic valve replacement in Medicare beneficiaries with BAV demonstrates 4 important findings. First, these data highlight that nearly one-third of

TABLE 3 Baseline Characteristic and Unadjusted Outcomes for Low-Risk Subgroup Aged <75 Years

Variable	Total (n = 7124)	TAVR (n = 1731)	SAVR (n = 5393)	*P*
Age, y	69.1 ± 3.01	70 ± 3.08	68.8 ± 2.93	<.001
Male sex	4654 (65.3)	1014 (58.6)	3640 (67.5)	<.001
White race	6432 (90.3)	1547 (89.4)	4885 (90.6)	.020
Frailty index	0.15 ± 0.036	0.158 ± 0.042	0.147 ± 0.033	<.001
Elixhauser score	5.09 ± 1.85	4.68 ± 1.86	5.23 ± 1.83	<.001
Diabetes mellitus	1501 (21.1)	471 (27.2)	1030 (19.1)	<.001
Peripheral artery disease	581 (8.16)	197 (11.4)	384 (7.12)	<.001
Chronic lung disease	1192 (16.7)	379 (21.9)	813 (15.1)	<.001
Gastrointestinal bleeding, ulcer, reflux	1846 (25.9)	420 (24.3)	1426 (26.4)	.077
Liver disease	10 (0.14)	2 (0.116)	8 (0.148)	.983
Atrial fibrillation	1409 (19.8)	319 (18.4)	1090 (20.2)	.113
Coronary artery disease	3066 (43)	902 (52.1)	2164 (40.1)	<.001
Anemia	521 (7.31)	171 (9.88)	350 (6.49)	<.001
Cancer	256 (3.59)	109 (6.3)	147 (2.73)	<.001
Cerebrovascular accident	722 (1.0)	154 (0.9)	602 (1.1)	.101
Permanent pacemaker	164 (2.3)	53 (3.06)	111 (2.06)	.02
Heart failure	2309 (32.4)	1009 (58.3)	1300 (24.1)	<.001
Obesity	1737 (24.4)	440 (25.4)	1297 (24)	.262
Malnutrition	69 (0.969)	18 (1.04)	51 (0.946)	.836
Chronic renal disease	908 (12.7)	294 (17)	614 (11.4)	<.001
Surgical predicted risk of mortality, %	0.67 (0.45–1.01)	0.68 (0.46–0.98)	0.66 (0.45–1.03)	.614
Hospital				
Length of stay, d	6 (2–8)	2 (1–3)	6 (3–8)	<.001
Total cost, $	33,326 ± 20,364	37,554 ± 23,890	31,968 ± 18,898	<.001
Acute kidney injury	729 (10.2)	76 (4.39)	653 (12.1)	<.001
Stroke	105 (1.47)	27 (1.56)	78 (1.45)	.821
Bleeding	236 (3.31)	32 (1.85)	204 (3.78)	<.001
New pacemaker	465 (6.53)	114 (6.59)	81 (1.51)	.004
Vascular complication	24 (0.337)	8 (0.462)	16 (0.297)	.426
Death	32 (0.449)	3 (0.173)	29 (0.538)	.077
30 days				
Readmit	595 (8.35)	108 (6.24)	487 (9.03)	<.001
Readmit bleeding	11 (0.154)	1 (0.058)	10 (0.185)	.409
Readmit stroke	20 (0.281)	4 (0.231)	16 (0.297)	.851
Readmit heart failure	251 (3.52)	53 (3.06)	198 (3.67)	.262
Any death	31 (0.435)	12 (0.693)	19 (0.352)	.098
5 years				
Any death	254 (3.57)	83 (4.79)	171 (3.17)	.002
Readmit	1982 (27.8)	475 (27.4)	1507 (27.9)	.707
Readmit bleeding	35 (0.491)	1 (0.058)	38 (0.63)	.006
Readmit stroke	125 (1.75)	40 (2.31)	85 (1.58)	.045
Readmit heart failure	740 (10.4)	219 (12.7)	521 (9.66)	<.001
Valve reintervention	31 (0.435)	7 (0.404)	24 (0.445)	.989
Composite[a]	306 (4.3)	115 (6.64)	191 (3.54)	<.001

[a]Composite: death, valve reintervention, or stroke. Data are presented as n (%), mean ± SD, or median (quartile 1–quartile 3).

predominantly low-risk Medicare beneficiaries between the ages of 65 and 85 years with BAV undergoing isolated aortic valve replacement receive TAVR, including nearly one-quarter of very low-risk patients aged <75 years, despite a lack of prospective evidence.

Second, after risk-adjustment, TAVR was associated with a 23% higher 5-year risk of heart failure readmission, 31% higher risk of stroke, and 33% higher risk of the composite of death, stroke, or valve reintervention compared with SAVR, but no difference in the 5-year valve reintervention rate.

Third, after risk adjustment, in the lowest-risk cohort of BAV patients aged <75 years, TAVR was associated with 22% higher risk of stroke and 37% higher composite risk of death, stroke, or

valve reintervention compared with SAVR, but no difference in the 5-year valve reintervention rate.

Finally, these data highlight that BAV patients undergoing TAVR have a different perioperative and longitudinal outcome profile than observed in prior trials, and in this Medicare population, TAVR for BAV stenosis compares less favorably with SAVR. Collectively, these findings may help to provide clarity for heart team clinical decision-making in BAV patients.

Despite excellent initial procedural outcomes, early signals have emerged for the potential of increased late incidence of embolic stroke in the first 2 years after TAVR.[20] Patients with BAV stenosis are known to commonly have higher calcium burden, often involving the left ventricular outflow tract. The potential of increased periprocedural stroke in heavily calcified BAVs poses a unique challenge and potential risk in this population.[21,22] The present study of >7000 BAV-only patients suggest a higher risk of stroke associated with TAVR, not only perioperatively but also cumulatively through the 5 years after the procedure.

Importantly, although the data set limited our ability to directly assess the magnitude of stroke, the focused definition used was documentation of a new stroke that served as the primary diagnosis requiring hospital admission. These findings are corroborated by the recently published NOTION-2 trial with 100 BAV patients highlighting a risk of death or disabling stroke at 1 year of 6.1% in the TAVR and 2.0% in the surgery group (HR, 3.1; $P = .3$).[14] The current study, powered to more clearly define outcomes, provides potentially important data for the heart team as patients with BAV stenosis are being considered for TAVR or SAVR. Particularly for low-risk patients, a preference may exist for a more invasive initial approach to mitigate longitudinal complications of stroke.

Gated computed tomographic angiography has become routine to assess anatomy in patients with aortic valve disease requiring aortic valve replacement, with emerging evidence of favorable BAV phenotypes for TAVR.[21,22] However, the geometric differences in tricuspid and bicuspid aortic valves have been well documented, and this has been cited as a concern for increased risk of perivalvular leak (PVL).[13,22-24]

Furthermore, longitudinal echocardiographic follow-up in prior studies has highlighted the long-term detrimental effects of moderate or greater PVL on patient outcomes.[25] Frequently, a significant PVL manifests clinically by increased symptoms of heart failure that do not resolve and is related to increase left ventricular strain. Although we were unable to assess echocardiographic characteristics in the present study, TAVR was associated with a 23% higher rate of readmission for the primary indication of heart failure over the 5-year study period compared with SAVR in patients with BAV. A similar signal of concern for PVL was observed in NOTION-2, with moderate or greater aortic insufficiency rates of 9.1% for TAVR and 0% for SAVR in the BAV patients.

Patients with BAV generally present with the need for aortic valve replacement at a younger age compared with trileaflet aortic valves.[26] Additionally, recent data highlight an exponential growth in TAVR for patients aged <65 years despite recent guidelines with indications only >65 years and a lack of data on long-term

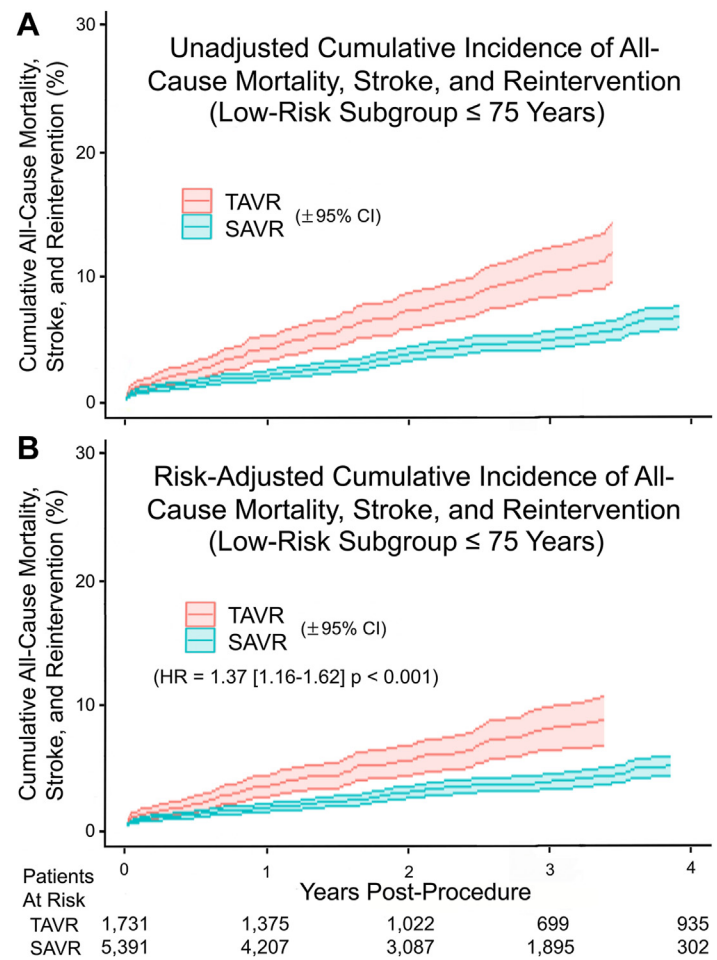

FIGURE 3 Cumulative incidence of all-cause mortality, stroke, and valve reintervention in low-risk subgroup aged ≤75 years. (A) Unadjusted cumulative incidence and (B) risk-adjusted cumulative incidence for composite primary outcome, SAVR vs TAVR with 95% CI (hazard ratio [HR], 1.37; $P < .001$).

TABLE 4 Risk-Adjusted Outcomes for Low-Risk Subgroup Aged <75 Years

Variable	Hazards Ratio for TAVR (95% CI)	*P* Value
Hospital		
AKI[a]	0.40 (0.35–0.46)	.001
Stroke[a]	1.25 (0.96–1.62)	.098
Bleeding[a]	1.02 (0.84–1.24)	.812
Vascular complication[a]	1.20 (0.66–2.18)	.540
New permanent pacemaker[a]	1.62 (1.43–1.84)	.001
Death[a]	1.15 (0.69–1.91)	.593
30 days		
Readmit[a]	0.56 (0.49–0.64)	<.001
Readmit bleeding[a]	0.13 (0.06–0.24)	.001
Readmit stroke[a]	1.12 (0.60–2.09)	.716
Readmit heart failure[a]	0.53 (0.43–0.64)	.001
5 years		
Death[b]	1.34 (0.94–1.89)	.097
Readmit[c]	1.02 (0.84–1.22)	.864
Readmit stroke[c]	1.22 (1.04–1.44)	.017
Readmit heart failure[c]	1.04 (0.67–1.59)	.874
Valve reintervention[c]	0.91 (0.64–1.27)	.808
Composite[c,d]	1.37 (1.16–1.62)	.001

[a]Multivariable logistic regression; [b]Cox proportional hazards model; [c]Fine-Gray multivariable time-to-event model accounting for competing risk of death; [d]Composite: death, valve reintervention, or stroke. TAVR, transcatheter aortic valve replacement.

durability.[12,27] Furthermore, a growing body of literature demonstrates a significantly higher risk of both surgical and transcatheter redo aortic valve replacement after TAVR, encouraging a thoughtful approach to the first intervention for BAV stenosis.[28-30] The combination of unfavorable early and late outcomes with TAVR in patients with BAV compared with SAVR, as well as the increased incidence of TAVR in these younger patients, who will require a much higher-risk redo valve in their lifetime, is currently disconcerting.

The present study, in combination with NOTION-2, raises important concerns about the use of TAVR in young low-risk patients with BAV. High-quality longitudinal data are needed to understand the safety and efficacy of TAVR in BAV. Multidisciplinary heart teams are faced with healthy BAV patients who want a minimally invasive procedure with short downtime, but these data suggest that TAVR may not be optimal in the BAV population. Given the current state, the time appears to be ripe for a well-designed pragmatic trial of SAVR vs TAVR in young and/or BAV patients with long-term follow-up to guide clinical management. In the absence of this future data, the current study may provide some data-driven equipoise when approaching these patients.

This study has several important limitations, including the generalizability of these findings for patients aged <65 years, which includes many BAV patients. Additionally, the administrative nature of the CMS database precludes a comparative evaluation of valve gradients or anatomy and the demonstration of causality. Despite doubly robust risk adjustment, the possibility of confounding remains, the effect of which, however, may be lessened by the large sample size and risk adjustment that included a comprehensive assessment of frailty.

Nevertheless, we were unable to account for patient and operating provider factors not captured in the database. Furthermore, concomitant aortic procedures were excluded from the analysis, but given the limited impact on longitudinal outcomes and potential coding bias for aneurysm being more common in surgical patients, this characteristic was not included in risk adjustment. Although the nature of the database precludes determination of timing of events during hospitalization, the CMS "present on admission" modifier assists in accurate categorization of predictor and outcome variables. Finally, we were unable to calculate STS risk score based on claims data but comprehensively adjusted using the CMS-based SAVR risk score, Elixhauser comorbidity index, Kim frailty index, and age.

Among Medicare beneficiaries with BAV undergoing isolated aortic valve replacement, SAVR was associated with higher index procedural morbidity and mortality but superior 5-year risk-adjusted freedom from all-cause mortality, stroke, heart failure, and composite of stroke, valve reintervention, or mortality compared with TAVR. Similar results were seen in the lowest-risk BAV patients aged <75 years, with TAVR being associated with significantly lower 5-year rates of longitudinal freedom from stroke and composite of death, stroke, or valve reintervention. These contemporary data may further inform heart team decision-making and future clinical trial design in patients with BAV.

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